

Personal Library & AI Research Engine – System Ontology & Semantic Knowledge Graph

W3C OWL 2 / RDF Standards: Dublin Core (dc:) • Bibliographic Ontology (bibo:) • W3C Provenance (prov-o:) • FOAF (foaf:)

Ontology v1.0.0 (TTL)

DOMAIN 1: SCHOLARLY ASSET & BIBLIOGRAPHIC ENTITIES (bibo:, dc:)

DocumentAsset

```
<<prov:Entity, owl:Class>>
+ guid: xsd:string [PK]
+ fileName: xsd:string
+ format: xsd:string
+ uploadDate: xsd:dateTime
```

BibliographicMetadata

```
<<bibo:Document, ValueObject>>
+ bibKey: xsd:string
+ entryType: xsd:string
+ title: xsd:string
+ doi: xsd:string [Optional]
```

TAXONOMY: ScholarlyPublication (subClassOf DocumentAsset)

AcademicArticle (bibo:Article)

BookPublication (bibo:Book)

ConferenceProceedings (bibo:Proc)

TechnicalReport (bibo:Report)

AuthorAgent

```
<<foaf:Person, prov:Agent>>
+ authoredBy relationship
```

PublisherAgent

```
<<foaf:Org, prov:Agent>>
+ publishedBy relationship
```

DOMAIN 2: DENSE VECTOR REPRESENTATION & QDRANT INDEXING

TextPassageChunk

```
<<prov:Entity, owl:Class>>
+ chunkIndex: xsd:integer
+ text: xsd:string (~500 chars)
+ pageNumber: xsd:integer
+ tokenCount: xsd:integer
```

:representedBy

DenseVectorEmbedding

```
<<owl:Class, Mathematical>>
+ dimensions: 768 (dense)
+ precision: float32
+ metric: CosineSimilarity
+ norm: L2-Normalized
```

VECTOR REASONING & EMBEDDING ENGINES

EmbeddingModel

```
<<prov:Agent, Transformer>>
+ Nomic Embed Text (8k ctx)
+ Fallback: text-embedding-004
```

VectorSpaceCollection

```
<<Qdrant HNSW Graph>>
+ personal_library_embeddings
+ M=16, efConstruct=100
```

:grounds (Evidence)

SemanticRetrievalQuery (k-NN / ANN)

Evaluates inner product: $\cos(\theta) = (A \cdot B) / (\|A\| \times \|B\|)$

Strictly filtered by :hasDocumentScope to isolate contexts

DOMAIN 3: DUAL-MODEL AI INFERENCE & MULTI-PERSPECTIVE SYNTHESIS

AnalyticalSummary

```
<<prov:Entity, owl:Class>>
+ summaryText: xsd:string
+ durationSeconds: xsd:decimal
+ generatedAt: xsd:dateTime
+ tokenCount: xsd:integer
```

GenerativeLLM

```
<<prov:Agent, FoundationModel>>
+ modelFamily: xsd:string
+ inferenceEngine: "Ollama"
+ contextWindow: xsd:integer
+ samplingTemp: xsd:float
```

DUAL-MODEL INFERENCE TAXONOMY

Llama3Model (subClassOf LLM)

Meta Llama 3.3 (70B Instruct)

temp = 0.2 (Strict Factuality)

→ ExecutiveTakeawaySummary

MistralLargeModel (subClassOf)

Mistral Large (2411)

temp = 0.3 (Fluency/Depth)

→ MethodologicalCritiqueSummary

Dual-Model Epistemic Triangulation

Contrasts independent model weights side-by-side to expose hallucinations

Records prompt tokens, completion tokens, and millisecond latencies

DOMAIN 4: CONVERSATIONAL RAG & GROUNDED CITATION EVIDENCE

ConversationalSession

```
<<prov:Activity, Session>>
+ sessionId: xsd:string [PK]
+ startTime: xsd:dateTime
+ userQuery: xsd:string
+ turnCount: xsd:integer
```

AssistantResponse

```
<<prov:Entity, Synthesized>>
+ answerText: xsd:string
+ responseTime: xsd:decimal
+ isGrounded: xsd:boolean = true
+ citationCount: xsd:integer
```

GroundedCitation

```
<<owl:Class, EvidenceLink>>
+ citationIndex: [1], [2]...
+ similarityScore: xsd:float
+ verbatimQuote: xsd:string
+ pageOffset: xsd:integer
```

PROVENANCE & EVIDENCE GROUNDING CHAIN (W3C PROV-O)

UserQuery → VectorRagService → ANN Top-K (Cosine ≥ 0.65) → Grounded Context

Every factual assertion in the AssistantResponse is bound to a GroundedCitation entity.

- :grounds (TextPassageChunk → GroundedCitation): Anchors citation to immutable chunk text.
- :citesEvidence (AssistantResponse → GroundedCitation): Attaches evidence tags to response text.
- :wasDerivedFrom (AssistantResponse → TextPassageChunk): W3C PROV-O data lineage tracking.

Cosine Threshold: ≥ 65%

Top-K Passages: K = 4 Chunks

Context Window: Zero Hallucination Guard

DOMAIN 5: GOVERNANCE, AUDIT LINEAGE & BPMN 2.0 WORKFLOW (PROV-O)

AuditSnapshot

```
<<prov:Entity, Immutable>>
+ snapshotGuid: xsd:string [PK]
+ versionNumber: xsd:integer
+ sha256Digest: xsd:string
+ savedAt: xsd:dateTime
```

BPMNWorkflowProcess

```
<<prov:Entity, Static BPMN 2.0 Export>>
+ processId: xsd:string
+ fileName: "Process_DocumentIngestionRAG"
+ purpose: External Camunda Modeler export
+ embeddedEngine: false
```

SecurityPrincipal

```
<<prov:Agent, Keycloak>>
+ subjectId: xsd:string [sub]
+ username: xsd:string
+ realm: "personal-library"
+ roles: Set<String> [RBAC]
```

IMMUTABLE REVISION EVOLUTION & NON-DESTRUCTIVE ROLLBACK

:hasHistoricalSnapshot (DocumentAsset → AuditSnapshot): Preserves past state

:previousVersionLineage (DocumentAsset → DocumentAsset): Self-referential graph

:governedBy (DocumentAsset → SecurityPrincipal): Role-Based Access Control

Non-Destructive Rollback Axiom:

Restoring version N does not prune intermediate states (> N). Snapshot N is cloned into a newly advanced version M = max(versions) + 1, creating an unbroken audit lineage.

SHA-256 Cryptographic Digest

Static BPMN Export: No Embedded Engine

RBAC: LIBRARY_ADMIN, CHIEF_RESEARCHER